

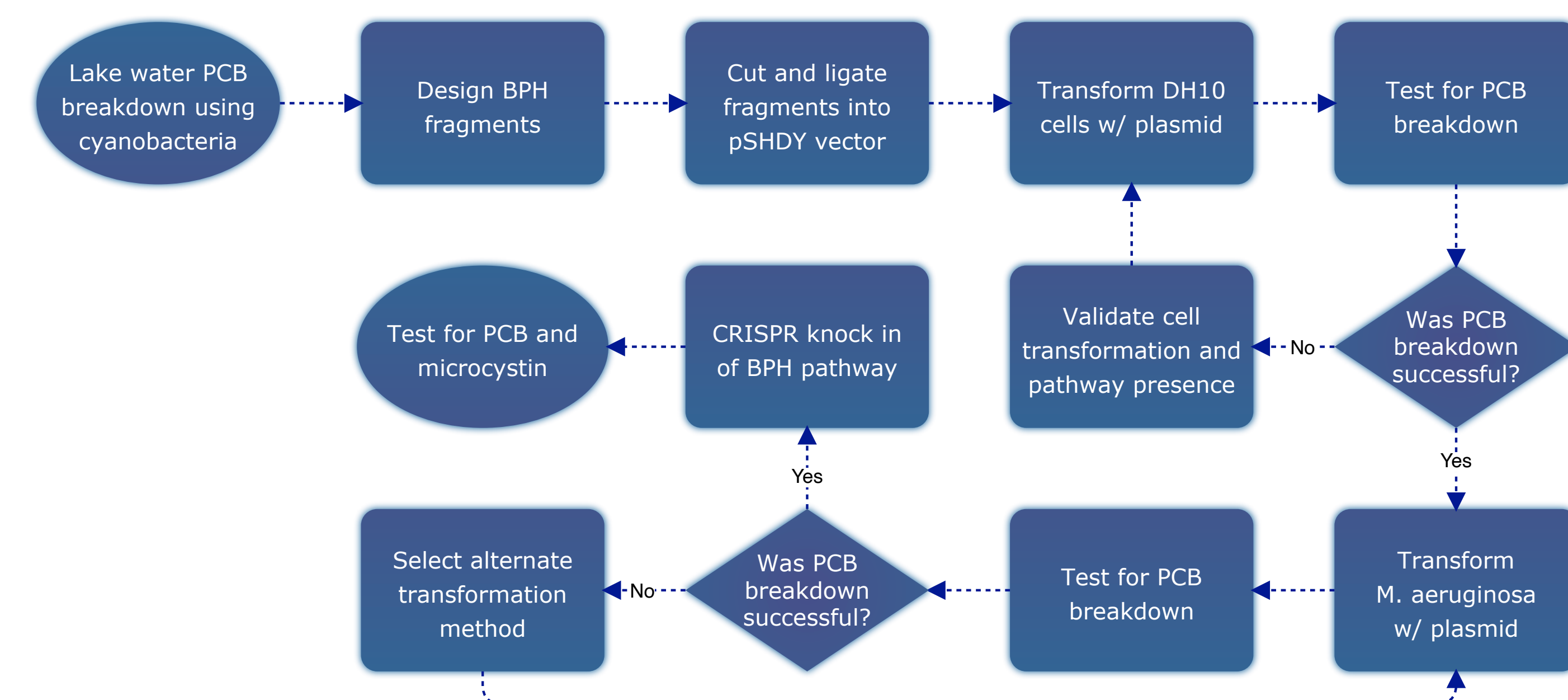
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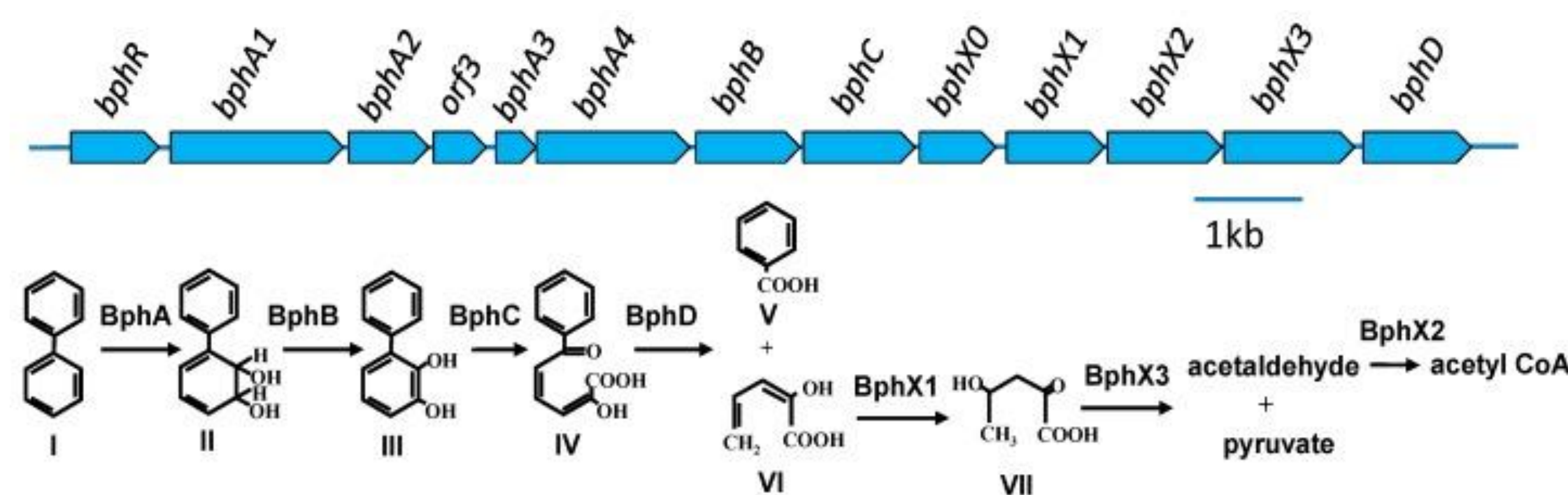
ABSTRACT

Polychlorinated Biphenyls (PCB) are aromatic hydrocarbon compounds that are highly carcinogenic and extremely stable. PCBs were once used in industrial products and can be found in the environment in soil and water. This project aims to utilize the BPH gene cluster by genetically engineering cyanobacteria to carry the BPH gene and provide a self-sustaining method of PCB removal. To accomplish this, the BPH genes have been synthesized and amplified with PCR then ligated and transformed into a bacterial host. The transformed E. coli is then tested in LB inoculated with PCBs before moving on to transforming cyanobacteria and testing water samples with PCBs. The final stage of this project would involve using CRISPR knock-in to replace a gene in Microcystis aeruginosa that produces the microcystin toxin, inhibiting toxin production and breaking down PCBs simultaneously.

BACKGROUND & OVERVIEW



PCB production was banned due to its negative impact on the environment; however, their presence persists due to their noted high stability. Bacterial strains that had been shown to break down PCBs were isolated from biphenyl-contaminated soil in Japan, then sequenced to provide a gene cluster (termed BPH).



METHODS & MATERIALS

The sequences for the BPH genes were synthesized by a third party. Due to the size of the gene cluster, the synthesized genes had to be received in 6 different plasmids, named BPH B, 1, 2, 3, 4, and E. The genes, along with the vector backbone, pSHDY, total a size of 20kb which require a bacterial strain that can handle a large size plasmid.



The completed plasmid named BpSHDY.



The BPH fragments were amplified using PCR. The amplified products were then ligated with the linearized vector using Gibson Assembly and transformed with the bacterial host dH10B.

Transformed cells were plated on LB with kanamycin and colonies were picked and grown overnight in LBkan media. The cultures were then purified, digested with enzymes, and ran on a 1% agarose gel to validate successful transformation.

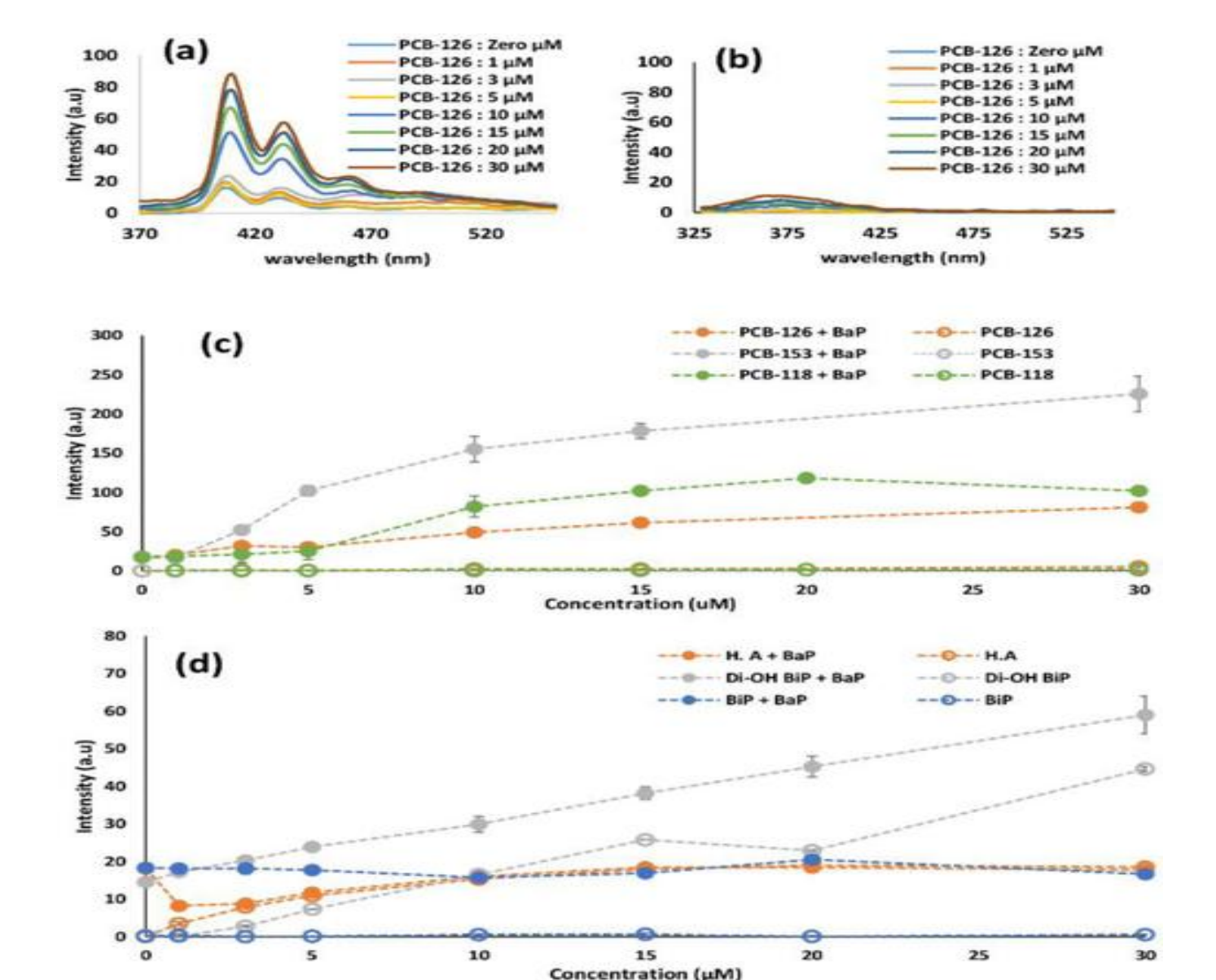


RESULTS

The results gathered thus far indicate an incomplete ligation as seen in the LBaa GA+ plate, the positive control for Gibson Assembly ligation. BpSHDY carries a resistance to kanamycin so the growth on the LBkan BpSHDY plate shows some promise but the growth shown on the negative control LBk NP (LBkan No Plasmid) hints towards growth of bacteria that is not the intended target. A plate of pGlo on LBamp+arabinose was additionally used as a positive control for successful transformation of dH10B cells. A double digest of a BpSHDY culture was performed resulting in no bands appearing at the expected sizes further confirming an unsuccessful ligation and transformation.

CONCLUSION & FUTURE WORK

Optimization of ligation and transformation must be performed before continuing forward. In this work, the pSHDY vector was linearized using restriction digest but due to low yield from gel extractions, a PCR product may provide more fruitful results. pSHDY also carries resistance to chloramphenicol and streptomycin so additional LB plates with the respective antibiotics may help narrow results to confirm a proper transformation.



Upon successful transformation of E. coli, cultures of LB will be inoculated with PCBs at different levels. Breakdown of PCB levels will then be measured using a fluorescence assay. After breakdown of PCBs using E. coli is confirmed, Microcystis aeruginosa will be the target for transformation with the BpSHDY plasmid.



REFERENCES & ACKNOWLEDGMENTS

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